Speech and Audio Systems

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Speech and Audio Systems

> Prepare Speech and Audio Systems: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/26, ai-3/12, ai-3/13. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt specification with observable response criteria

Reference example, not the only acceptable wording: Teacher explanation: Fields status,room,capacity,source_ids.

Status supported or needs_review. supported requires matching current approval and current capacity for the same room; no unresolved conflict.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

Concept to explain

Speech interaction includes **capture**, recognition, interpretation, and response stages.

Term	Meaning
capture	Record a signal, observation or item of information for later use.
speech recognition	Converting spoken audio into an estimated sequence of words.
intent	The goal a user is trying to achieve, which may need clarification.
synthesis	Combining information into a new account or output while preserving its meaning and support.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can separate **speech recognition**, calculation and speech **output** when diagnosing an error.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Speech and Audio Systems using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Learner says ‘define erosion’ in a noisy room.

2. Recognizer **output**s ‘design erosion.’

3. **Intent** stage follows the wrong transcript.

4. Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Exact explanation to model

> **Why the result follows**
>
> **Transcription** supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

Explain the added worked example

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech

Term	Meaning
Error propagation	An earlier error affecting later processing

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Speech and Audio Systems

A fictional learner dictates a calculation to a voice assistant.

The facts we will use

Supplied facts

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Supplied facts

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

AI ACADEMY / AI-3/12

Speech and Audio Systems

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recognise speech

Transcription changes thirteen into thirty.



2. Calculate text

30+7=37 is correct for the mistaken transcript.



3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



4. Confirm intended input

Read back the recognised expression; intended 13+7=20.

CHANGE ONE THING

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Recognise speech	Transcription changes thirteen into thirty.
2. Calculate text	30+7=37 is correct for the mistaken transcript.
3. Speak result	Speech synthesis repeats 37; fluent voice does not validate the input.
4. Confirm intended input	Read back the recognised expression; intended 13+7=20.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

What this example does not establish

Keep the stated limit

These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

Change one thing in the pictured case

Change one thing

Change: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Predicted result

Prediction: The calculator and spoken result should become 20 under the declared pipeline.

Why the result follows

Reason: 13+7=20; the changed **input** removes the earlier recognition error.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Speech and Audio Systems, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A speech tool hears fifteen as fifty, then calculates correctly using fifty.

2. State the decision to explain

Write the question to explain: Where did the error enter?

3. Reveal the worked reasoning

Reveal after **prediction**s: **Transcription** supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

Hold back the independent task

Keep for an independent attempt: A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage	Teacher action
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

Reasoned teacher answer

Speech recognition was wrong. The calculator correctly computed 30+7; **synthesis** correctly repeated that wrong-for-**intent** result.

Changed-case reasoning: The calculator and spoken result should become 20 under the declared pipeline. 13+7=20; the changed **input** removes the earlier recognition error.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Speech and Audio Systems investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Trace four audio failures stage by stage, locate the first mismatch, and propose a stage-specific test or fallback.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Speech and Audio Systems questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong **input**.

Independent question

Independent question · withhold the key

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

Expected independent answer

Teacher key · Expected independent answer

The **transcription** stage introduced the error. Compare the transcript with the recording before trusting or summarizing the number.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A summary can repair an upstream **transcription** error without checking the audio.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

The transcript changed sixteen to sixty. Display or read back the interpreted numbers before calculation when ambiguity matters. The intended result is 4; the calculator correctly computed its incorrect **input**.

Retain only necessary diagnostic information.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its

input, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

W2 Trace the worked case

Workbook question · W2

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

Teacher reasoning · W4

Teacher explanation: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: Known audio says “nineteen minus six”. **Transcription** says “ninety minus six”; the calculator returns84. Locate the first error and propose a check.

Selected case answer

Teacher key · Selected case answer

The first error is **transcription**:19 became90. The calculator correctly computes90–6=84 for its received **input**, but the intended result is13. Display or read back the recognized expression before acting.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Speech and Audio Systems to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Show transcripts and require confirmation before saving memory or using tools.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 13: Multimodal **Evidence**. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said. Require intermediate reasoning and the stated scope.

Fresh retry: Transcript says eight; a validated order record says eighteen and is declared authoritative for ordering.

Expected reasoning: Use eighteen for the bounded order-record task; retain the transcript discrepancy and do not claim perfect **transcription**.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Retrieval and indexing. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/13, ai-3/15, ai-3/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Speech and Audio Systems lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is

unsupported.

Check every learner before discussion: Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Correct the transcript to “thirteen plus seven” with all later stages unchanged. Require a **prediction** and a reason before comparing.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

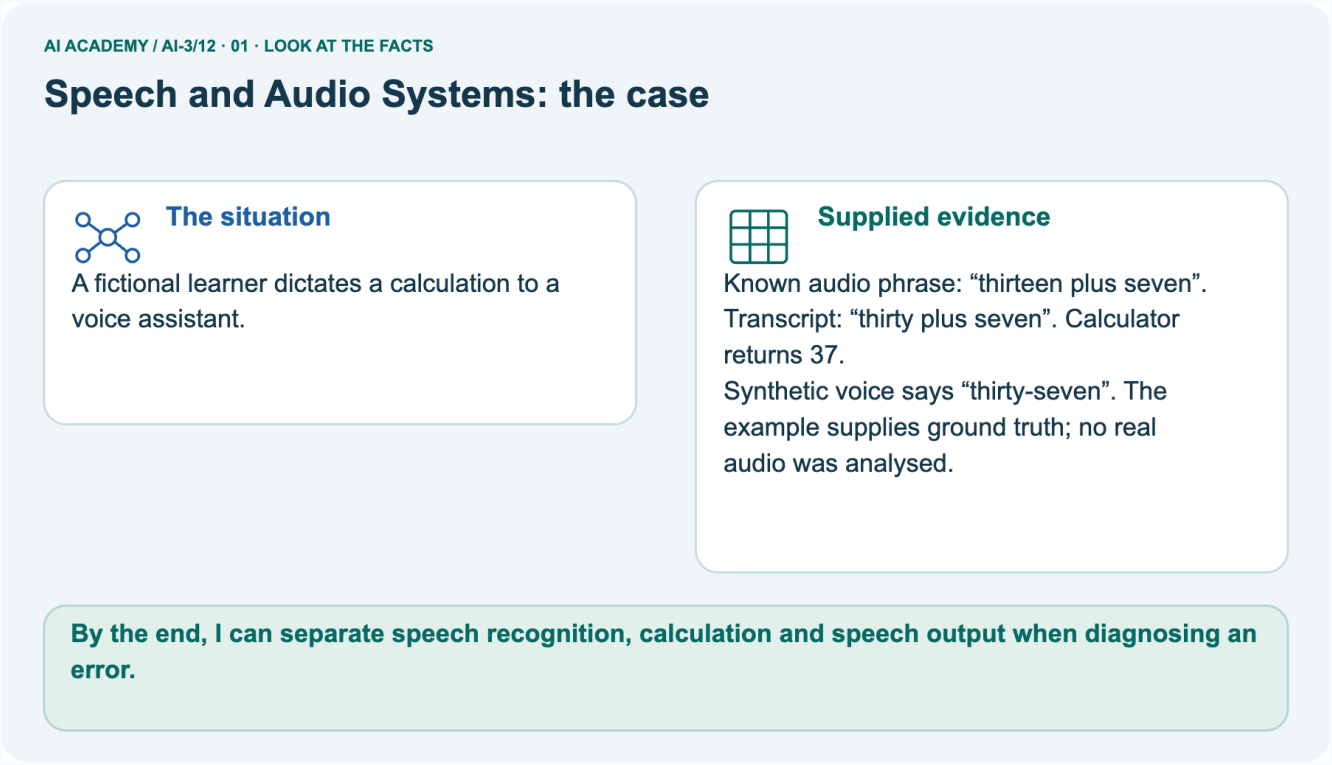
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Speech and Audio Systems graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Speech and Audio Systems: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional learner dictates a calculation to a voice assistant. Supplied facts: Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37. Synthetic voice says “thirty-seven”.

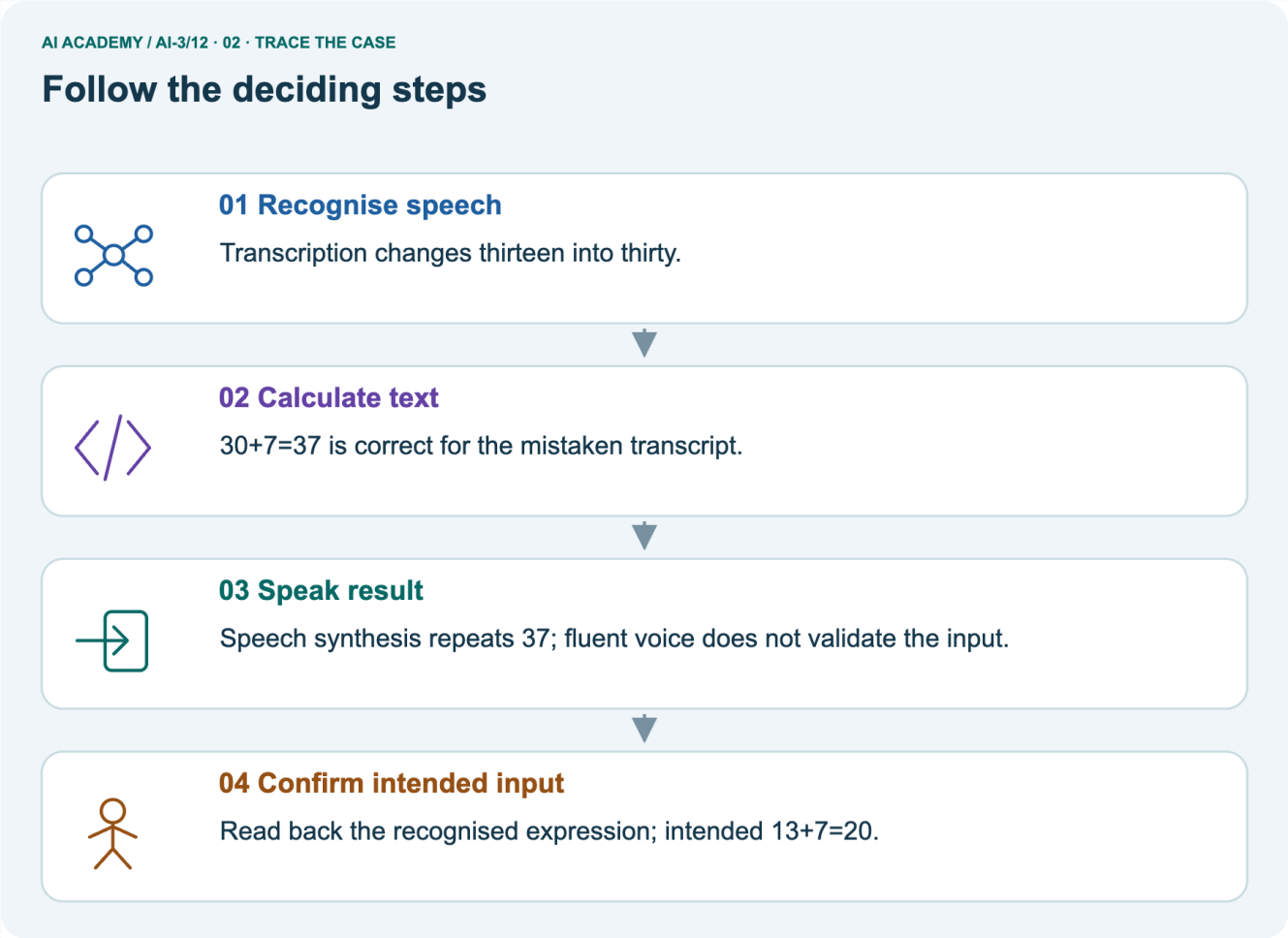
The example supplies ground truth; no real audio was analysed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

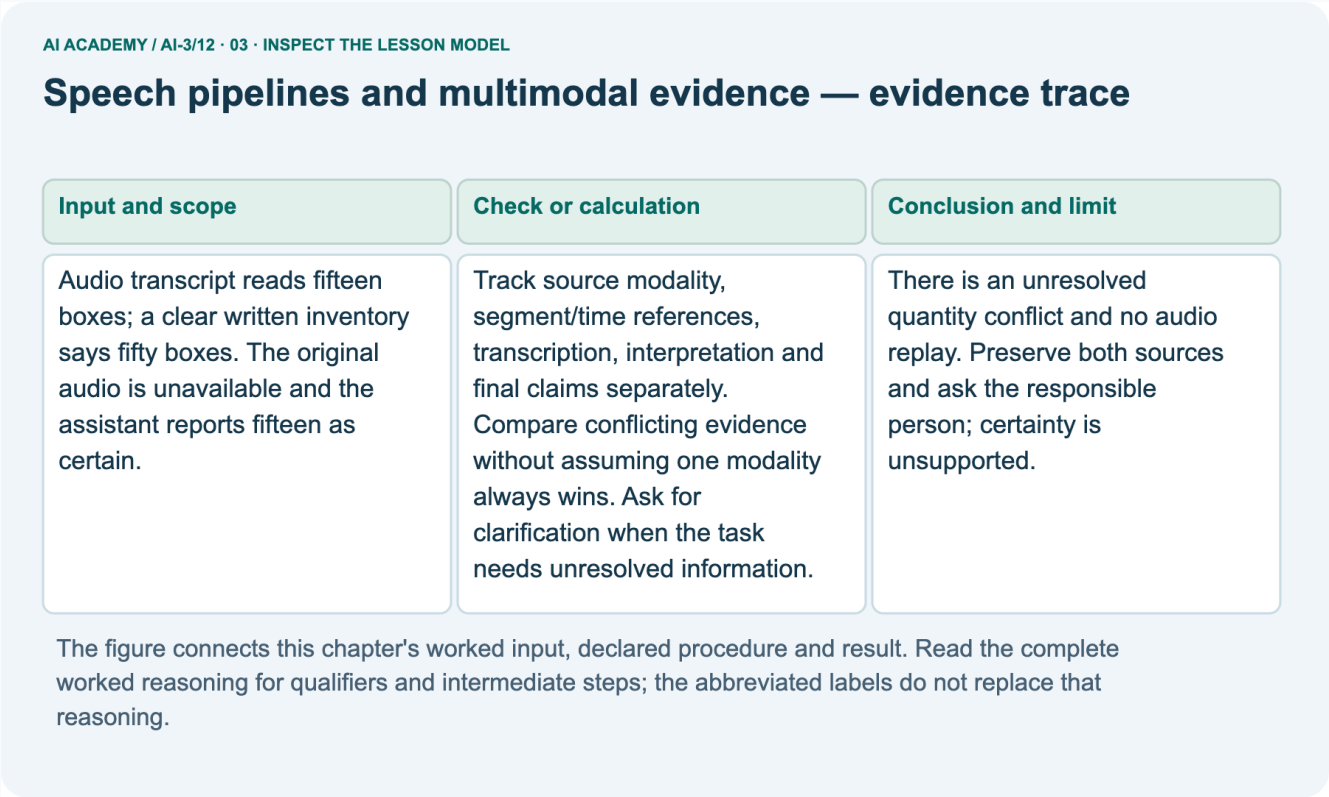
Recognise speech: **Transcription** changes thirteen into thirty. → Calculate text: 30+7=37 is correct for the mistaken transcript. → Speak result: Speech **synthesis** repeats 37; fluent voice does not validate the **input**. → Confirm intended **input**: Read back the recognised expression; intended 13+7=20.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

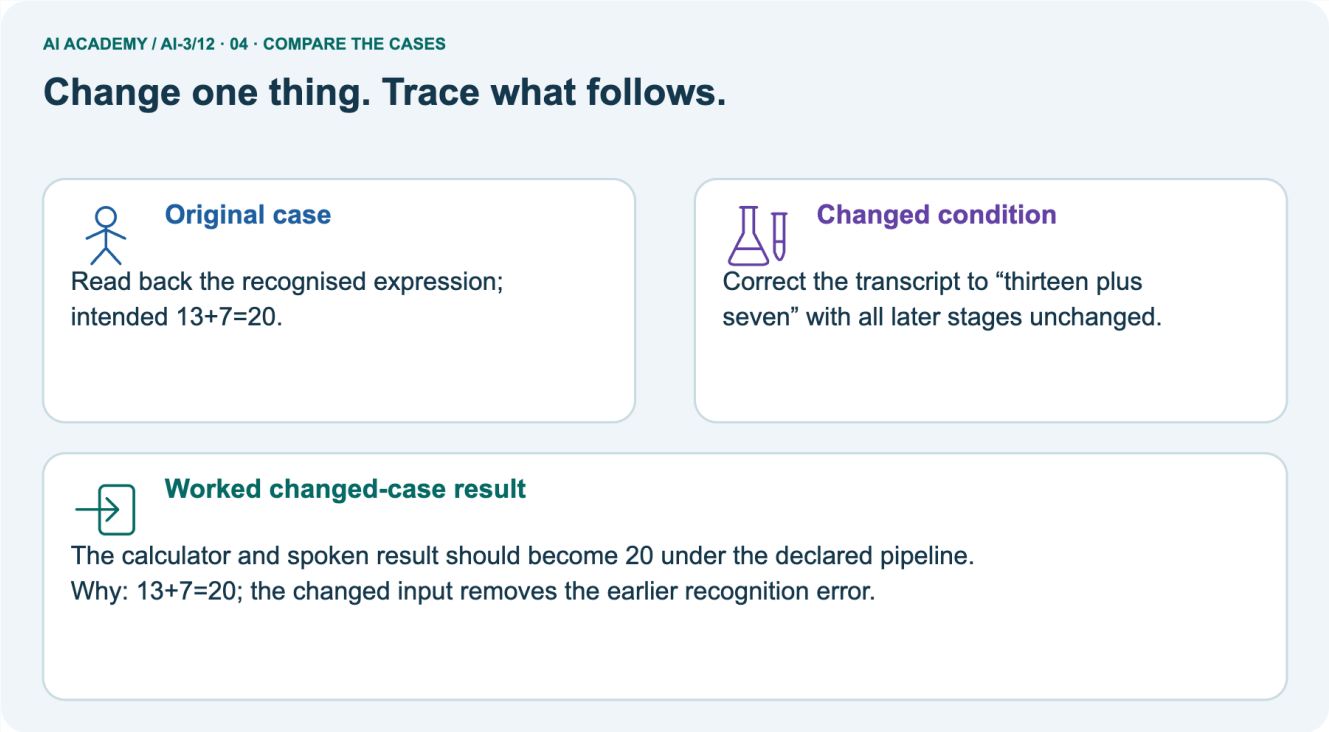
Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Read back the recognised expression; intended 13+7=20. Changed condition: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Expected result: The calculator and spoken result should become 20 under the declared pipeline. Reason: 13+7=20; the changed **input** removes the earlier recognition error.

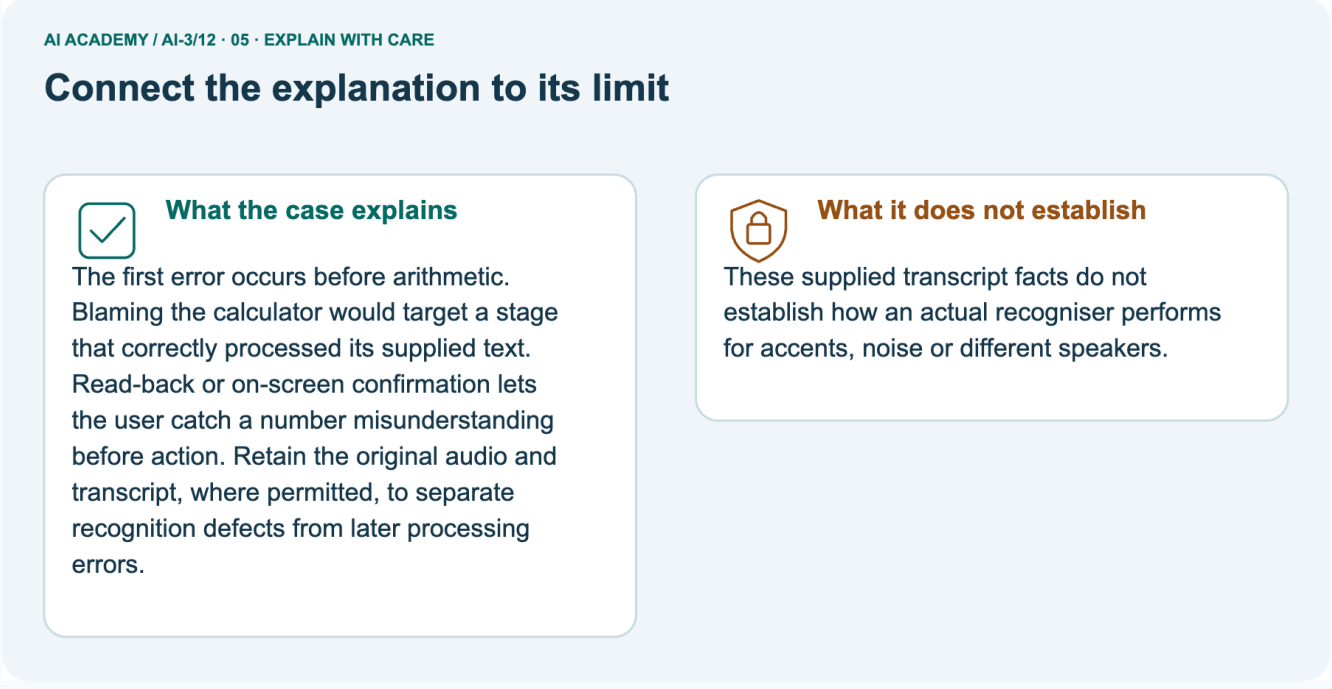
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 13+7=20; the changed **input** removes the earlier recognition error.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly **process**ed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action.

Retain the original audio and transcript, where permitted, to separate recognition defects from later **process**ing errors.

Limit: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

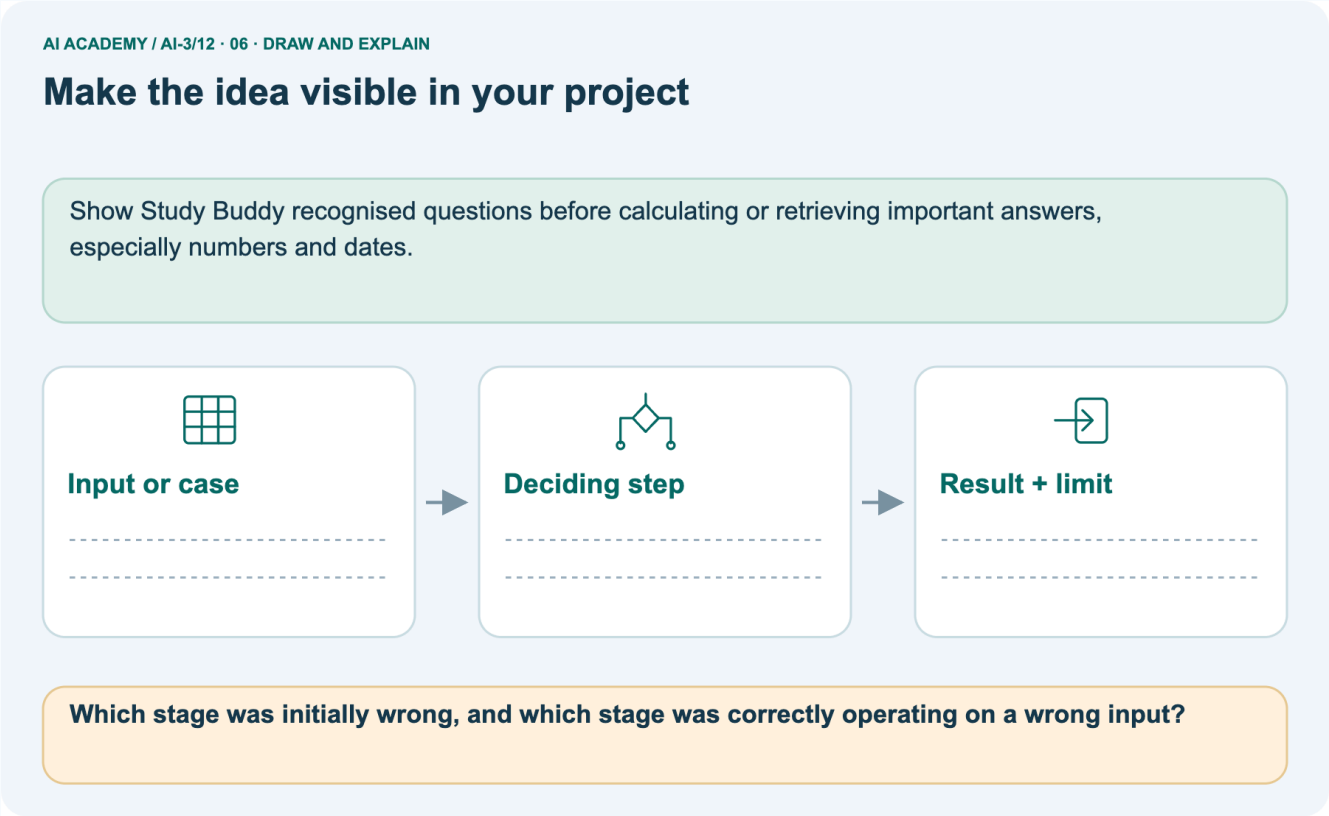
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates. Draw your **input**, deciding step, result and limit.

Supported question: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: **Speech recognition** was wrong. The calculator correctly computed 30+7; **synthesis** correctly repeated that wrong-for-**intent** result.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.